

SAURABH KUMAR

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SUMMARY

Full-stack developer and BCA graduate with hands-on experience building and deploying production-grade AI applications. Designed and shipped a load-tested music streaming platform on AWS, including a LangGraph-based AI agent microservice with retrieval-augmented generation (RAG). Seeking full-stack or GenAI engineering roles at product and AI-first companies.

TECHNICAL SKILLS

Languages: TypeScript, Python, SQL, C++

Backend: Node.js, Express.js, FastAPI, REST APIs, Socket.IO

AI / ML: LangGraph, Gemini 3.1, RAG, pgvector

Databases: PostgreSQL, MongoDB, Redis

Cloud & DevOps: AWS (EC2, S3, RDS, CloudFront), Docker, GitHub Actions CI/CD

Frontend: React, Next.js

Tools: Git, Linux, VS Code

EXPERIENCE

Freelance Web Developer | Patna, Bihar

2026

- Designed and developed a responsive website for Gyanodaya Public School, Patna using Next.js, React, and Tailwind CSS.
- Implemented SEO best practices and optimized page performance; deployed on Vercel with a custom domain.
- Managed the project independently end-to-end, from client requirements gathering to deployment and handover.

PROJECTS

Sonix Music — AI-Powered Music Streaming Platform

TypeScript, Node.js, React, PostgreSQL, Redis, BullMQ, Docker, AWS (EC2, S3, RDS, CloudFront), GitHub Actions

Live Demo: <https://sonix.saurabhx.site> | **Source Code:** github.com/Saurabh-singhx/sonix_music_app

- Built a production-grade full-stack music streaming platform with secure authentication (Google OAuth and OTP) and presigned AWS S3 URLs for media storage.
- Developed REST APIs in Node.js and TypeScript with Redis caching and BullMQ background job queues for asynchronous workloads.
- Containerized the backend with Docker and deployed on AWS EC2, with the frontend served via S3 and CloudFront on a custom domain, using GitHub Actions CI/CD.
- Load-tested the platform to 200-500 concurrent users on a single t3.micro EC2 instance using k6.

Sonix AI Agent — LangGraph Recommendation Microservice

Python, FastAPI, LangGraph, Gemini 3.1, pgvector, PostgreSQL

Source Code: [Saurabh-singhx/sonix_recommendation_ai_agent](https://github.com/Saurabh-singhx/sonix_recommendation_ai_agent)

- Architected a standalone Python FastAPI microservice using LangGraph to power AI-driven music recommendations within Sonix Music.
- Implemented tool-calling workflows with Gemini 3.1, including structured output generation and tool-vs-node error handling patterns.
- Built a retrieval-augmented generation (RAG) pipeline using pgvector for semantic similarity search over music metadata.
- Integrated the agent with the core Node.js backend via async SQLAlchemy and a shared Supabase PostgreSQL instance.

EDUCATION

Bachelor of Computer Applications (BCA)

T.P.S. College, Patna, Patliputra University, Bihar — July 2022 to July 2025

ACHIEVEMENTS

- Solved 350+ data structures and algorithms problems on LeetCode.
- Load-tested Sonix Music API to 200-500 concurrent users using k6 on a single AWS EC2 t3.micro instance.